CIS 6270 Practice Exam 1 for Exam 2

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Follow these instructions for a timed practice attempt:

- Please write your name on each page. Allow yourself 90 minutes.
- Complete your first attempt individually before consulting the solutions. Record questions to discuss afterward with the instructor and teaching assistants.
- Calculators are not permitted. Every part is multiple choice with six options. Work in the space provided, then box exactly one option letter for every part. When grading, count only the boxed letter.
- Each labeled part is worth 0.5 points and receives either 0 or 0.5 points. There is no partial credit.
- You may use a one-page cheat sheet (8.5in $\times$ 11in, one side only) containing only material provided in this class. Do not use a calculator, phone, or other device during the timed attempt. Keep your signed cheat sheet with your completed work.

The University of Pennsylvania is a community dedicated to scholarship, leadership, and service and to the principles of honesty, fairness, respect, and accountability. I will use this practice exam to assess my own understanding. I will complete the timed attempt using my own work before consulting the solutions or discussing answers.

This practice exam is worth 12 points and does not contribute to your course grade.

Name: _____

Signature: _____

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This practice exam is worth 12 points.

Beating Autoregression on LM1B with Staggered Block Diffusion

You lead a group that trains language models. As you know from Lecture 4, an autoregressive transformer still has the lowest perplexity among the models we studied on LM1B, a benchmark built from English news sentences. With 327B training tokens, a retrained transformer had a test perplexity of 20.86, and MDLM had a perplexity bound of 23.00. In the BD3-LM comparison, which used the same training budget for every model, the transformer had a test perplexity of 22.83, while MDLM and BD3-LM with blocks of four had bounds of 31.78 and 28.23. As its blocks shrink, BD3-LM approaches autoregression, yet on LM1B, every discrete diffusion model in the lecture fell short of the transformer.

In block diffusion, we keep the chain rule between blocks and fill each block by masked diffusion. Blocks of one token recover autoregression, but then we generate only one token per network call. In this practice exam, you will design a new block diffusion model, which we call staggered block diffusion. We keep the blocks of four from BD3-LM and mask later positions sooner than earlier ones. During generation, we therefore tend to fill each block from left to right. During training, we still corrupt every block at random noise levels.

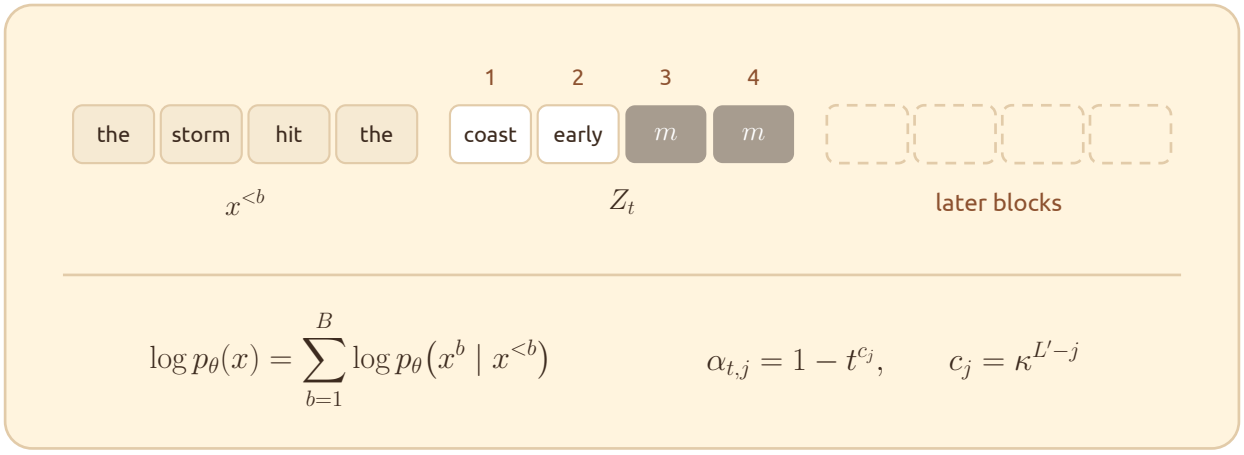


Figure 1. Staggered block diffusion on LM1B. We split each sentence into blocks of $L' = 4$ tokens and generate the blocks in order, each given the clean blocks before it. Inside the current block, offset j keeps its token with probability $\alpha_{t,j} = 1 - t^{c_j}$, with larger exponents c_j at earlier offsets. Corruption runs from the clean block at $t = 0$ to the all-mask block at $t = 1$, and generation runs back from $t = 1$ to $t = 0$.

Figure 1 outlines the construction. You will follow the steps from the end of Lecture 4 for building a new discrete diffusion model. You will choose its rate matrix, compute its forward marginal, derive its reverse rates and loss, sample from it, and steer it toward a topic with Doob's h-transform. You will then interpret hypothetical LM1B results for this model.

Research brief and notation

Each training sentence $x = (x_1, \ldots, x_L)$ has $L = 128$ tokens from the LM1B vocabulary $\mathcal{V}$, which contains $K = 30{,}522$ tokens, and $m \notin \mathcal{V}$ denotes the mask. We split x into $B = L/L'$ blocks of L' tokens. We write x^b for block b, $x^{<b}$ for the clean blocks before it, and x^b_j for the token at offset $j \in \{1, \ldots, L'\}$ of block b. As in BD3-LM, the chain rule holds between blocks:

$$\log p_\theta(x) = \sum_{b=1}^{B} \log p_\theta\big(x^b \mid x^{<b}\big).$$

We model each factor with masked diffusion on the current block, given the clean prefix. Time t runs from $t = 0$, where the block is clean, to $t = 1$, where every offset is masked. The noisy block is $Z_t = (Z_{t,1}, \ldots, Z_{t,L'})$, and z denotes one of its values. Every offset keeps its token independently, with survival probability

$$\alpha_{t,j} = 1 - t^{c_j}, \qquad c_j = \kappa^{L'-j}, \qquad \kappa \geq 1.$$

With $\kappa = 1$, every offset follows MDLM's linear schedule $\alpha_t = 1 - t$, which recovers BD3-LM with that schedule. With $\kappa > 1$, earlier offsets survive longer. Our main runs use $L' = 4$ and $\kappa = 2$, so $(c_1, c_2, c_3, c_4) = (8, 4, 2, 1)$.

The denoiser $d^j_\theta(a \mid z, x^{<b})$ is the probability that offset j of the current block holds the clean token $a \in \mathcal{V}$, given the noisy block z and the clean prefix. We write $z^{j \leftarrow a}$ for z with offset j set to a. All rates, marginals, and posteriors below condition on the prefix $x^{<b}$. Perplexity is the exponential of the average negative log-likelihood per token in nats, and log denotes the natural logarithm.

Step	**Points**
1. Locate the gap to autoregression	1.5
2. Choose the staggered rate matrix	2.0
3. Derive the reverse rates and the loss	2.5
4. Sample a block	2.0
5. Guide generation toward a topic	2.0
6. Interpret hypothetical LM1B results	2.0
Total: 24 parts	**12.0**

Step 1: Locating the Gap to Autoregression (1.5 points)

Before we change the model, we need to read the LM1B numbers correctly and connect the block factorization to autoregression.

(a) We know only bounds on the perplexity of the diffusion models, while we know the transformer's perplexity exactly. We should first check what a bound allows us to conclude.

In the BD3-LM comparison, the transformer had a test perplexity of 22.83, BD3-LM with blocks of four had a bound of 28.23, and MDLM had a bound of 31.78. Which conclusion do these three numbers support? Choose ONE.

A. BD3-LM's test perplexity exceeds the transformer's by exactly 5.40.

B. BD3-LM's test perplexity is at least 28.23, so it is worse than the transformer's.

C. BD3-LM's test perplexity is at most 28.23, so it may lie above or below 22.83.

D. MDLM's test perplexity exceeds BD3-LM's, since 31.78 is larger than 28.23.

E. BD3-LM's test perplexity lies somewhere between 22.83 and 28.23.

F. The transformer's 22.83 is also a bound, so no ranking of the three models follows.

(0.5 points)

(b) By the chain rule, the probability of a sentence is a product of block probabilities. We need to convert that product into the perplexity that we report on LM1B.

A model splits a sentence of six tokens into three blocks of two tokens and assigns the blocks probabilities 1/2, 1/4, and 1/8, each given its prefix. What is the perplexity of this sentence? Choose ONE.

A. 2

B. 4

C. 8

D. 64

E. 14/3

F. $\sqrt{2}$

(0.5 points)

(c) In Lecture 4, we noted that blocks of one token recover autoregression. We can check this directly from the block loss.

Take $L' = 1$ with MDLM's linear schedule $\alpha_t = 1 - t$, so every block contains one token x^b and its loss is

$$\int_0^1 \frac{1}{t} \Pr(Z_t = m) \left(-\log d_\theta(x^b \mid m, x^{<b})\right) \mathrm{d}t.$$

The denoiser does not receive t, so its prediction is the same at every time when the token is masked. What does this loss equal? Choose ONE.

A. $\frac{1}{2}\big(-\log d_\theta(x^b \mid m, x^{<b})\big)$

B. $2\big(-\log d_\theta(x^b \mid m, x^{<b})\big)$

C. It diverges, because $\int_0^1 \mathrm{d}t/t$ is infinite.

D. $-\log d_\theta(x^b \mid m, x^{<b}) + \log K$

E. $-\log d_\theta(x^b \mid m, x^{<b})$

F. $-\log d_\theta(x^b \mid m, x^{<b}) - 1 + d_\theta(x^b \mid m, x^{<b})$

(0.5 points)

Step 2: Choosing the Staggered Rate Matrix (2 points)

In GenMD4, each token had its own masking rate. We can similarly assign a rate to each offset so that later offsets are masked sooner. We then need to confirm that these rates satisfy the first three facts from Lecture 4.

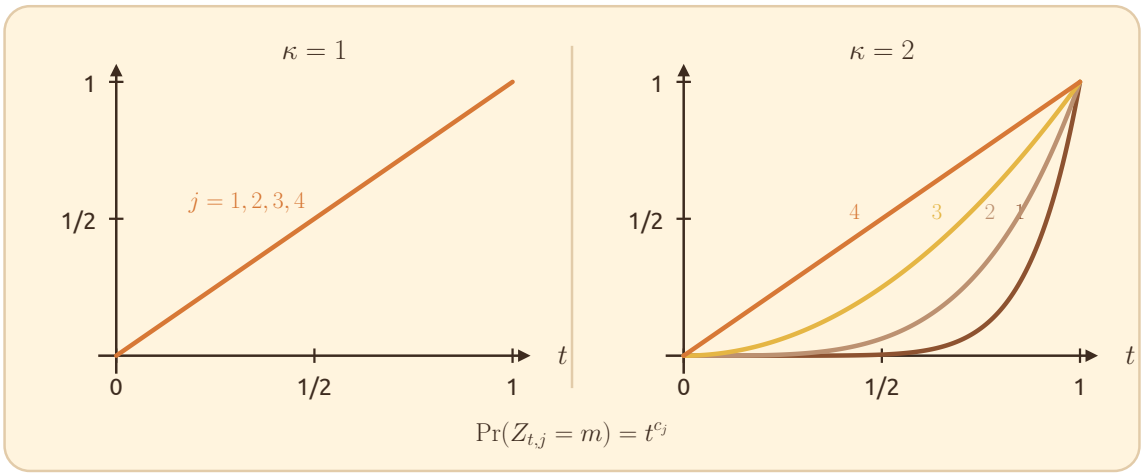


Figure 2. Masking probability $\Pr(Z_{t,j} = m) = t^{c_j}$ at each offset j of a block of $L' = 4$. With $\kappa = 1$, all four offsets follow the linear schedule. With $\kappa = 2$, the exponents are $(c_1, c_2, c_3, c_4) = (8, 4, 2, 1)$, and offset 4 is masked soonest.

(a) To write the rate matrix, we first need the masking rate corresponding to our survival schedule.

A clean token at offset j jumps to m at rate $\beta_j(t)$, and m never jumps back. Which rate $\beta_j(t)$ produces the survival schedule $\alpha_{t,j} = 1 - t^{c_j}$? Choose ONE.

A. $\dfrac{c_j}{t}$

B. $-\log\left(1 - t^{c_j}\right)$

C. $c_j\, t^{c_j - 1}$

D. $\dfrac{c_j\, t^{c_j - 1}}{1 - t^{c_j}}$

E. $\dfrac{c_j\, t^{c_j}}{1 - t^{c_j}}$

F. $\dfrac{c_j}{1 - t}$

(0.5 points)

(b) Using this rate, we can track how probability flows into the mask at one offset.

Let $p_t(m) = \Pr(Z_{t,j} = m)$. Which equation is the master equation at the mask state? Choose ONE.

A. $\dot{p}_t(m) = \beta_j(t)\, p_t(m)$

B. $\dot{p}_t(m) = \beta_j(t)\big(1 - p_t(m)\big)$

C. $\dot{p}_t(m) = \beta_j(t)\big(1 - p_t(m)\big) - \beta_j(t)\, p_t(m)$

D. $\dot{p}_t(m) = -\beta_j(t)\big(1 - p_t(m)\big)$

E. $\dot{p}_t(m) = \dfrac{c_j}{t}\big(1 - p_t(m)\big)$

F. $\dot{p}_t(m) = \beta_j(t)$

(0.5 points)

(c) We can also solve the master equation in closed form, which gives the second fact.

At offset j, the rate matrix is $R_t = \beta_j(t) R_\pi$ with $R_\pi = \mathbf{1}\pi^\top - I$ and $\pi = \delta_m$. Define the clock $\tau_j(t) = \int_0^t \beta_j(v)\,\mathrm{d}v$. Which transition matrix $P_{0,t}$ gives the forward marginal of offset j? Here u denotes the uniform distribution over $\mathcal{V}$. Choose ONE.

A. $e^{-\tau_j(t)} I + \big(1 - e^{-\tau_j(t)}\big)\mathbf{1}\delta_m^\top$ with $\tau_j(t) = -\log\big(1 - t^{c_j}\big)$

B. $e^{-\tau_j(t)} I + \big(1 - e^{-\tau_j(t)}\big)\mathbf{1}\delta_m^\top$ with $\tau_j(t) = c_j\, t$

C. $e^{-\tau_j(t)} I + \big(1 - e^{-\tau_j(t)}\big)\mathbf{1}u^\top$ with $\tau_j(t) = c_j\, t$

D. $\big(1 - e^{-\tau_j(t)}\big) I + e^{-\tau_j(t)}\mathbf{1}\delta_m^\top$ with $\tau_j(t) = c_j\, t$

E. $e^{\tau_j(t)} I + \big(1 - e^{\tau_j(t)}\big)\mathbf{1}\delta_m^\top$ with $\tau_j(t) = -\log\big(1 - t^{c_j}\big)$

F. $e^{-\tau_j(t)} I + \big(1 - e^{-\tau_j(t)}\big)\mathbf{1}\delta_m^\top$ with $\tau_j(t) = t^{c_j}$

(0.5 points)

(d) In Step 4, we will need the probability that a visible token becomes masked between two times. Survival to a later time requires survival to the earlier time first.

For $s < t$, what is the probability that offset j is masked at time t, given that it holds its clean token a at time s? Choose ONE.

A. $\dfrac{t^{c_j} - s^{c_j}}{1 - s^{c_j}}$

B. $\dfrac{t^{c_j} - s^{c_j}}{t^{c_j}}$

C. $\dfrac{1 - t^{c_j}}{1 - s^{c_j}}$

D. $t^{c_j} - s^{c_j}$

E. $1 - e^{-c_j(t-s)}$

F. $(t-s)^{c_j}$

(0.5 points)

Step 3: Deriving the Reverse Rates and the Loss (2.5 points)

The fourth and fifth facts concern generation and training. We need reverse rates that come from the posterior over clean tokens, and a loss that bounds the likelihood and is minimized at the true reverse rates.

(a) At a masked offset, we need to know how fast to reveal a token and which token to reveal. In Lecture 4, we reversed a jump with a ratio of probabilities and then averaged over the clean token with the denoiser.

Write d for $d_\theta^j(a \mid z, x^{<b})$. Which pair gives the conditional reverse rate $\hat{R}_t(m \to a \mid x^b)$ at a masked offset j and the model rate $\hat{R}_t^\theta(m \to a)$? Choose ONE.

A. $\big(\beta_j(t)\,\delta(a, x_j^b),\quad \beta_j(t)\,d\big)$

B. $\big(\frac{c_j}{t}\,\delta(a, x_j^b),\quad \beta_j(t)\,d\big)$

C. $\big(\frac{1}{t}\,\delta(a, x_j^b),\quad \frac{1}{t}\,d\big)$

D. $\big(c_j\,t^{c_j-1}\,\delta(a, x_j^b),\quad c_j\,t^{c_j-1}\,d\big)$

E. $\big(\frac{c_j}{t}\,\delta(a, x_j^b),\quad \frac{c_j}{t}\,d\big)$

F. $\big(\beta_j(t)\,\delta(a, x_j^b),\quad \frac{c_j}{t}\,d\big)$

(0.5 points)

(b) To train the denoiser, we compare these model rates with the conditional targets through the rate error $\ell(A, B) = A \log(A/B) - A + B$.

Sum the rate error over a, then over offsets and blocks. The expectation is over sentences x and noisy blocks Z_t drawn from their forward marginals. Here $Z_t^{<b}$ denotes the earlier blocks corrupted to the same time. Which expression is the staggered block loss? Choose ONE.

A. $\displaystyle\sum_{b=1}^{B} \int_0^1 \mathbb{E}\Big[\sum_{j:\, Z_{t,j}=m} \frac{1}{t}\big(-\log d_\theta^j(x_j^b \mid Z_t, x^{<b})\big)\Big]\, \mathrm{d}t$

B. $\displaystyle\sum_{b=1}^{B} \int_0^1 \mathbb{E}\Big[\sum_{j:\, Z_{t,j}=m} \frac{1}{t}\big(-\log d_\theta^j(x_j^b \mid Z_t, Z_t^{<b})\big)\Big]\, \mathrm{d}t$

C. $\displaystyle\sum_{b=1}^{B} \int_0^1 \mathbb{E}\Big[\sum_{j:\, Z_{t,j}=m} \frac{c_j}{t}\big(-\log d_\theta^j(x_j^b \mid Z_t, x^{<b}) - 1 + d_\theta^j(x_j^b \mid Z_t, x^{<b})\big)\Big]\, \mathrm{d}t$

D. $\displaystyle\sum_{b=1}^{B} \int_0^1 \mathbb{E}\Big[\sum_{j:\, Z_{t,j}=m} \frac{1}{t}\big(-\log d_\theta^j(x_j^b \mid Z_t, x^{<b}) - 1 + d_\theta^j(x_j^b \mid Z_t, x^{<b})\big)\Big]\, \mathrm{d}t$

E. $\displaystyle\sum_{b=1}^{B} \int_0^1 \mathbb{E}\Big[\sum_{j:\, Z_{t,j}=m} \frac{c_j}{t}\big(-\log d_\theta^j(x_j^b \mid Z_t, x^{<b})\big)\Big]\, \mathrm{d}t$

F. $\displaystyle\sum_{b=1}^{B} \int_0^1 \mathbb{E}\Big[\sum_{j:\, Z_{t,j}=m} \frac{c_j}{t}\big(-\log d_\theta^j(x_j^b \mid Z_t, Z_t^{<b})\big)\Big]\, \mathrm{d}t$

(0.5 points)

(c) In SEDD, the network predicts the probability ratios in the reverse rate directly. We should check how these ratios relate to our denoiser.

At a masked offset j, we wrote the ratio in Lecture 4 as

$$\frac{p_t(z^{j\leftarrow a})}{p_t(z)} = \sum_{y\in\mathcal{V}} \Pr\big(x_j^b = y \mid Z_t = z\big)\, \frac{q_t(a \mid y)}{q_t(m \mid y)},$$

where $q_t(\cdot \mid y)$ is the forward marginal of offset j with clean token y. Write $d^j(a) = \Pr(x_j^b = a \mid Z_t = z)$ for the exact posterior. Which expression does this ratio equal for our schedule? Choose ONE.

A. $\dfrac{t^{c_j}}{1-t^{c_j}}\, d^j(a)$

B. $\dfrac{c_j}{t}\, d^j(a)$

C. $d^j(a)$

D. $\beta_j(t)\, d^j(a)$

E. $\big(1-t^{c_j}\big)\, d^j(a)$

F. $\dfrac{1-t^{c_j}}{t^{c_j}}\, d^j(a)$

(0.5 points)

(d) During training, the integrand of the loss in (b) sums the weighted cross-entropies of the masked offsets.

At time $t = 1/2$, offsets 2 and 4 of a block are masked, and offsets 1 and 3 are visible. The denoiser's cross-entropies at the clean tokens are 1.0 at offset 2 and 0.5 at offset 4. With $(c_1, c_2, c_3, c_4) = (8, 4, 2, 1)$, what is the contribution of these two offsets to the loss integrand? Choose ONE.

A. 3

B. 9/2

C. 9

D. 12

E. 1

F. 23/15

(0.5 points)

(e) Many clean sentences are consistent with the same noisy block, so the conditional targets in (b) change from one training example to the next.

Fix the noisy block z and the prefix $x^{<b}$. Over the clean blocks consistent with z, the conditional target rate toward a at a masked offset j is $A = \frac{c_j}{t}\,\delta(a, x_j^b)$. Which model rate B minimizes the expected rate error $\mathbb{E}[\ell(A, B)]$ over the training sentences with prefix $x^{<b}$ whose current block is corrupted to z? Choose ONE.

A. $\frac{c_j}{t}\,\delta(a, \hat{x}_j)$, where $\hat{x}_j$ is the most probable clean token at offset j

B. $\frac{c_j}{t}\,\Pr(x_j^b = a \mid x^{<b})$, the probability given the prefix alone

C. $\beta_j(t)\,\Pr(x_j^b = a \mid x^{<b})$

D. $\frac{c_j}{t}\,\Pr(x_j^b = a \mid Z_t = z, x^{<b})$

E. No single rate minimizes it, because A changes with the clean block.

F. $\frac{c_j}{t}\big(1 - \Pr(x_j^b = a \mid Z_t = z, x^{<b})\big)$

(0.5 points)

Step 4: Sampling a Block (2 points)

With a trained denoiser, we generate the blocks in order. We initialize each block with all masks and run its reverse process from $t = 1$ to $t = 0$, conditioned on the clean prefix. We then append the completed block to the prefix (Figure 3).

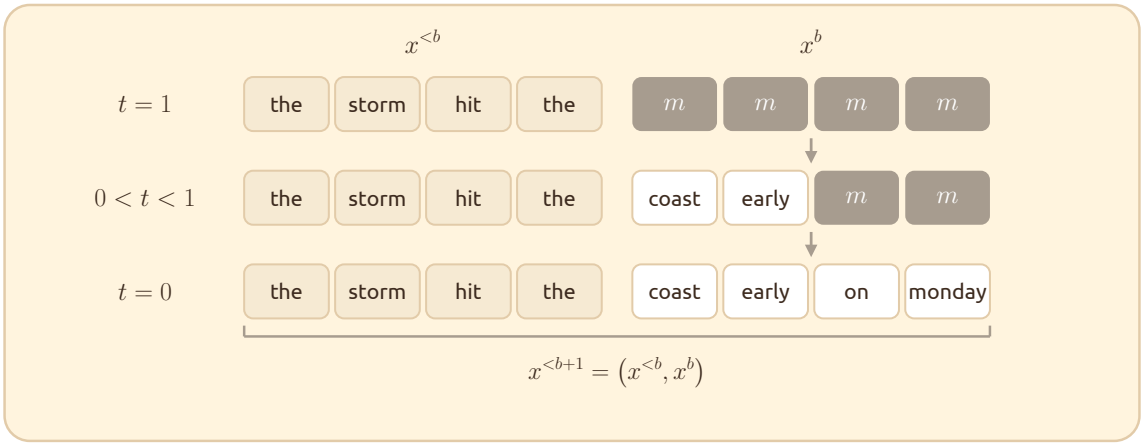


Figure 3. Generating one block of $L' = 4$ tokens. Given the clean prefix $x^{<b}$, the current block starts from all masks at $t = 1$ and is clean at $t = 0$. A reverse step can reveal several offsets, and a revealed token stays fixed. The completed block then joins the prefix of block $b + 1$.

(a) Our first reverse step starts from the all-mask block, and each offset becomes visible independently.

Take one exact posterior step from $t = 1$ to $s = 1/2$ with $(c_1, c_2, c_3, c_4) = (8, 4, 2, 1)$. Which vector lists the probabilities that offsets 1, 2, 3, and 4 become visible in this step? Choose ONE.

A. $(1/2,\ 3/4,\ 15/16,\ 255/256)$

B. $(255/256,\ 15/16,\ 3/4,\ 1/2)$

C. $(1/256,\ 1/16,\ 1/4,\ 1/2)$

D. $(1/2,\ 1/2,\ 1/2,\ 1/2)$

E. $(1,\ 1,\ 1,\ 1/2)$

F. $(8/15,\ 4/15,\ 2/15,\ 1/15)$

(0.5 points)

(b) Later in generation, we can save model calls with first-order τ-leaping. However, every probability in a step must stay valid.

At $t = 1/2$, all four offsets of a block are still masked. In a first-order step of length h, offset j becomes token a with probability $h\,\hat{R}_t^\theta(m \to a)$ and stays masked with the remaining probability. What is the largest h for which every probability in this step is valid? Choose ONE.

A. $1/2$

B. $1/30$

C. $1/8$

D. $1/4$

E. $1/64$

F. $1/16$

(0.5 points)

(c) In P2, we revisit earlier choices using a corrector at a fixed time. Under its rate matrix $C_t = R_t + \hat{R}_t$, the marginal p_t remains stationary.

Write d for $d_\theta^j(a \mid z, x^{<b})$. At time t, which pair gives the rate of remasking a visible token at offset j and the rate of revealing token a at a masked offset j under C_t? Choose ONE.

A. $\big(\beta_j(t),\ \ \frac{c_j}{t}\, d\big)$

B. $\big(\frac{c_j}{t},\ \ \beta_j(t)\, d\big)$

C. $\big(\beta_j(t),\ \ \beta_j(t)\, d\big)$

D. $\big(\frac{c_j}{t},\ \ \frac{c_j}{t}\, d\big)$

E. $\big(0,\ \ \frac{c_j}{t}\, d\big)$

F. $\big(\beta_j(t) + \frac{c_j}{t},\ \ \frac{c_j}{t}\, d\big)$

(0.5 points)

(d) Each reverse step costs one network call, since we cache the clean prefix. Fewer steps per block therefore save time.

Suppose we take a single exact posterior step per block, from $t = 1$ to $s = 0$, and write $m^{L'}$ for the all-mask block. From which distribution do we draw the block x^b? Choose ONE.

A. $\prod_{j=1}^{L'} d_\theta^j\big(x_j^b \mid x_1^b, \ldots, x_{j-1}^b, x^{<b}\big)$, each offset given the offsets before it

B. $\prod_{j=1}^{L'} d_\theta^j\big(x_j^b \mid x_{-j}^b, x^{<b}\big)$, each offset given the other offsets of the block

C. $d_\theta^1\big(x_1^b \mid m^{L'}, x^{<b}\big)$ at offset 1, with the other offsets still masked

D. $\prod_{j=1}^{L'} d_\theta^j\big(x_j^b \mid m^{L'}, x^{<b}\big)$, each offset given only the prefix

E. The data distribution of the block given the prefix, since every reveal probability is correct

F. Each offset becomes visible with probability $1 - 2^{-c_j}$ and is drawn from $d_\theta^j(\cdot \mid m^{L'}, x^{<b})$

(0.5 points)

Step 5: Guiding Generation toward a Topic (2 points)

So far, we have generated LM1B sentences without controlling their topic. We now want to generate sentences about a chosen topic without retraining the denoiser. As in Lecture 4, we can condition the reverse process on an event E with Doob's h-transform. Here E is the event that a topic classifier labels the finished sentence as sports (Figure 4).

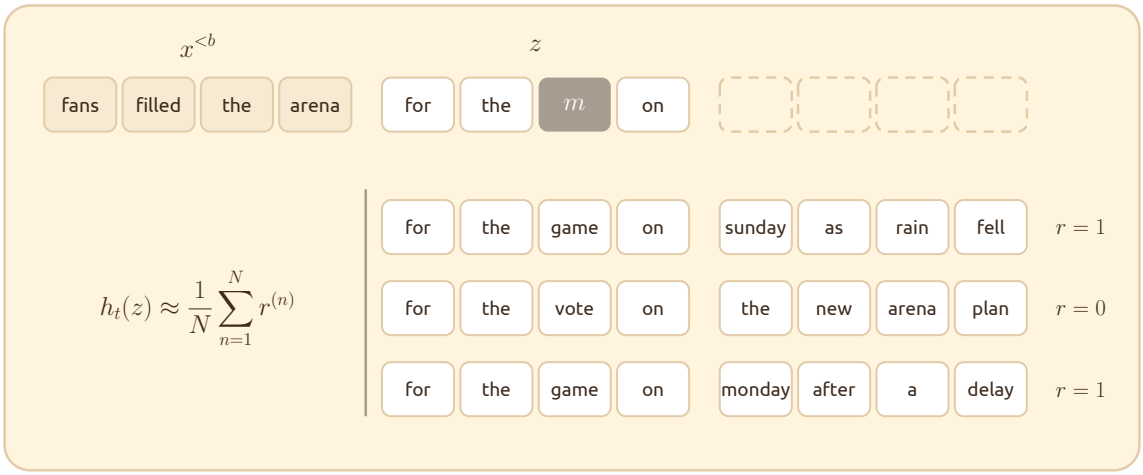


Figure 4. Guiding one block toward a topic. The event E depends on the finished sentence, so the value $h_t(z) = \Pr(E \mid Z_t = z, x^{<b})$ averages over the masked offsets of the current block and over every later block. We estimate it by completing the sentence with rollouts and scoring each completion.

(a) The event concerns the finished sentence, and we generate that sentence one block at a time.

Let $h_t(z) = \Pr(E \mid Z_t = z, x^{<b})$ for the current block, and write d for $d_\theta^j(a \mid z, x^{<b})$. Which expression is the guided rate of revealing token a at a masked offset j? Choose ONE.

A. $\dfrac{c_j}{t}\, d\, \dfrac{h_t(z)}{h_t(z^{j\leftarrow a})}$

B. $\dfrac{c_j}{t}\, d\, h_t(z^{j\leftarrow a})$

C. $\dfrac{c_j}{t}\, d\, \dfrac{h_t(z^{j\leftarrow a})}{h_t(z)}$

D. $\beta_j(t)\, d\, \dfrac{h_t(z)}{h_t(z^{j\leftarrow a})}$

E. $\dfrac{c_j}{t}\, d\, \dfrac{\Pr(E \mid x^{<b})}{h_t(z)}$

F. $\dfrac{c_j}{t}\, d + h_t(z^{j\leftarrow a}) - h_t(z)$

(0.5 points)

(b) At $t = 1/2$, offset 3 of the current block is masked, and $c_3 = 2$.

The denoiser assigns probabilities 1/2, 1/4, and 1/4 on the words *game*, *vote*, and *film*, and zero probability to all other words. A classifier gives $h_t(z) = 0.4$ for the current block, and $h_t(z^{3\leftarrow a}) = 0.6$, 0.2, and 0.2 after revealing *game*, *vote*, and *film*. Which guided rates toward (*game*, *vote*, *film*) does the h-transform give? Choose ONE.

A. $(2,\ 1,\ 1)$

B. $(4/3,\ 2,\ 2)$

C. $(6/5,\ 1/5,\ 1/5)$

D. $(3/4,\ 1/8,\ 1/8)$

E. $(3,\ 1/2,\ 1/2)$

F. $(12/5,\ 4/5,\ 4/5)$

(0.5 points)

(c) A classifier on noisy blocks can be hard to train. In PepTune, the value is estimated by completing the sequence with the denoiser and averaging the rewards of the completions.

We estimate the values at offset 3 the same way. Each completion fills the rest of the current block and every later block, and it earns reward 1 if the classifier labels the finished sentence as sports and 0 otherwise. The completions after revealing *game* earn $(1, 1, 0, 1)$, and those after revealing *vote* earn $(0, 1, 0, 0)$. The current value estimate is 0.5, and the unguided rates toward *game* and *vote* are 2 and 1. Given that the next reveal at offset 3 is *game* or *vote*, what is the probability that it is *game*? Choose ONE.

A. 2/3

B. 3/4

C. 2/5

D. 11/16

E. 3/5

F. 6/7

(0.5 points)

(d) We repeat rollouts for every sample. In TR2-D2, we retain the search results and use them to fine-tune the model toward a reward-tilted target. We can then sample without rollouts.

Two completions of a prefix have model probabilities $3/4$ for a sentence about politics and $1/4$ for a sentence about sports, with rewards 0 and $\log 3$. The target is $p^\star(x) \propto p_\theta(x)\exp\big(r(x)/\lambda\big)$. Which target probabilities of (politics, sports) does temperature $\lambda = 1/2$ give? Choose ONE.

A. $(1/2,\ 1/2)$

B. $(1/4,\ 3/4)$

C. $(3/4,\ 1/4)$

D. $(27/28,\ 1/28)$

E. $(1/10,\ 9/10)$

F. $(0,\ 1)$

(0.5 points)

Step 6: Interpreting Hypothetical LM1B Results (2 points)

We trained staggered block diffusion on LM1B with $L' = 4$ at the BD3-LM budget, for several values of κ. We will interpret the hypothetical results in Table 1 and Figure 5 using the facts derived above.

Model	κ	Training perplexity	Test perplexity
Transformer	–	18.42	22.83
BD3-LM	1	$\leq$ 27.41	$\leq$ 28.23
Staggered	2	$\leq$ 22.90	$\leq$ 24.10
Staggered	4	$\leq$ 20.31	$\leq$ 21.95
Staggered	16	$\leq$ 19.07	$\leq$ 22.41
Staggered	256	$\leq$ 18.61	$\leq$ 22.79

Table 1. Hypothetical LM1B perplexities at the BD3-LM budget with blocks of four. The transformer's values are exact, and the diffusion values are bounds.

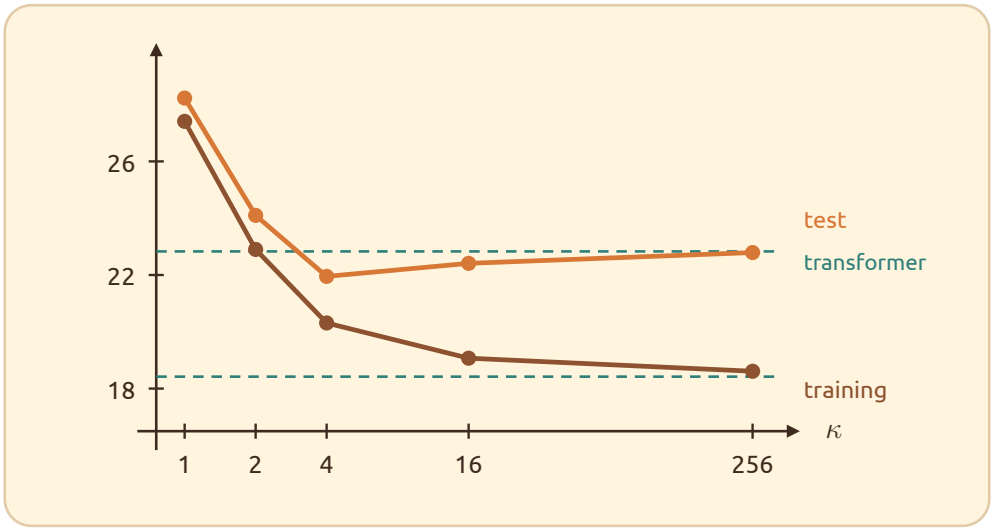


Figure 5. The values of Table 1 against κ on a logarithmic axis. Dashed lines mark the transformer's training and test perplexities.

(a) With $\kappa = 4$, we reached the lowest test value in Table 1.

Which conclusion does its bound of 21.95 support, compared with the transformer's 22.83? Choose ONE.

A. Its test perplexity is at most 21.95, so it is below the transformer's 22.83.

B. Nothing follows about the comparison, because a bound on perplexity may be loose.

C. Its test perplexity is at least 21.95, so it may still exceed 22.83.

D. It beats the transformer only if its bound turns out to be tight.

E. A bound and an exact perplexity cannot be compared in either direction.

F. Its test perplexity is exactly 21.95, because the staggered bound is tight.

(0.5 points)

(b) In Table 1, we also list each model's perplexity on its training sentences.

Which interpretation of the κ sweep agrees with both columns? Choose ONE.

A. The bound loosens at every offset as κ grows, so the training and test values both rise with κ.

B. Larger κ adds a positive prior term at $t = 1$, which raises the test value and leaves training unchanged.

C. Larger κ raises the weights c_j/t, so we reach a lower loss on the training sentences and the test sentences alike.

D. Beyond $\kappa = 4$, the training value keeps falling while the test value rises, which is the pattern of overfitting.

E. With larger κ, we need fewer network calls, which lowers the training value and raises the test value.

F. The test value must fall as κ grows, so its rise after $\kappa = 4$ reflects noise in the evaluation.

(0.5 points)

(c) In Run E, we repeated the run with $\kappa = 4$, using MDLM's weight $1/t$ for every masked offset in the loss. The test value computed with this weight was 2.8.

What can we conclude from this number? Choose ONE.

A. It is a valid bound and a tighter one, because $1/t \leq c_j/t$ at every offset.

B. It is a valid but looser bound, because the weight $1/t$ is the same at every offset.

C. It may not bound the test perplexity, because $1/t \leq c_j/t$ and the smaller weight shrinks the masked terms.

D. It is the exact test perplexity, because $1/t$ is the reveal weight of the linear schedule.

E. It is a valid bound, because the forward process still ends at the all-mask block.

F. It may not bound the test perplexity, because $1/t \geq c_j/t$ and the larger weight inflates the masked terms.

(0.5 points)

(d) In Run F, we repeated the run with $\kappa = 4$ and also supplied the time t to the denoiser, as in the code from Lecture 4. We obtained the same test bound of 21.95.

Which fact explains why adding the time input did not improve the bound? Choose ONE.

A. The weight c_j/t already brings t into the loss, so a time input to the network is redundant.

B. At a given time, every offset of the block has the same masking probability, which removes the stagger from the posterior.

C. The prefix is clean at every time, so time has no effect on the denoiser's prediction.

D. By Bayes' rule, the posterior depends on t through $\alpha_{t,j}$, and the number of masks in z determines t.

E. In Bayes' rule, the factors $\alpha_{t,j}$ and $1 - \alpha_{t,j}$ are the same for every clean block that agrees with z, so they cancel.

F. The weights c_j/t are the same at every offset, so the time cancels from the loss.

(0.5 points)

End of practice exam.

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